

Simon Yan

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Education

University of California, Santa Barbara
Bachelor of Science in Computer Science

Sep 2023 – Mar 2027
Goleta, CA

Technical Skills

Languages: Python, Java, JavaScript, TypeScript, Go, C++, C, SQL, HTML/CSS
Frameworks & Libraries: React, Next.js, Node.js, FastAPI, Flask, PyTorch, TensorFlow, LangChain
Tools & Technologies: AWS, Docker, Git, Linux, REST APIs, gRPC, CI/CD, PostgreSQL, Supabase
Relevant Coursework: Data Structures & Algorithms, Object-Oriented Design, Deep Learning, Generative AI, Databases, Operating Systems, Distributed Systems, Networks

Experience

Software Engineering Intern

Shopify

Sep 2026 – Present

Bellevue, WA

- Improve Shopify's merchant business forecasting foundation model evaluation scores by **4.5%** on downstream prediction tasks by compressing admin browsing activity into session-level summaries.
- Design BigQuery preprocessing to consolidate **578 admin page events into 18 categories**, reducing histories exceeding the model context limit by **99.98%**.

Software Development Engineer Intern

Amazon

Jun 2026 – Sep 2026

Seattle, WA

- Built a configuration store using **AWS DynamoDB and Lambda** for Amazon's catalog ML enrichment pipeline, eliminating manual code deployments for configuration changes by migrating hard-coded values across multiple packages into NoSQL tables.
- Developed a management interface with audit trails and instant rollback by implementing two-person approvals, versioned snapshots, and concurrency locking, **reducing configuration updates from hours to seconds**.
- Reduced end-to-end pipeline time for catalog backfill jobs by over **120x**, from 2.6 days P90 to under 30 minutes per product.

AI/ML Software Engineering Intern

Censys

Jun 2025 – Sep 2025

Ann Arbor, MI

- Developed a **multi-agent summarization service** with **8 specialized LLM agents** to parallelize analysis of large-scale internet host data, reducing context overhead and improving summary quality.
- Built an **LLM-as-a-judge evaluation framework** to compare generated summaries on factual consistency, coverage, relevance, and conciseness against source data.
- Designed a file directory interface for internet hosts, simplifying malicious file identification, and **led user acceptance testing** to improve usability based on feedback.

Project Lead and Officer

ACM Industry

Sep 2025 – Present

Goleta, CA

- Lead **company-sponsored projects** by defining project scope and guiding student development teams.
- Present **interactive workshops** on industry topics including database design and applied AI for **more than 100 students**.

Projects

Launchpad | *Go, Docker, AWS, PostgreSQL, gRPC*

Sep 2026 – Present

- Built an **AI agent-native** deployment runtime that enables AI coding agents to package applications into containers and return standalone, shareable URLs without local setup.
- Developed a **distributed container orchestration** layer with resource scheduling, dynamic routing, failure recovery, scale-to-zero, and TTL-based cleanup across a multi-node worker cluster.

RTX RF Signal Classifier | *C++, RISC-V, SIMD, BLAS, SLEEF*

Dec 2025 – Jun 2026

- Built a C++ tensor framework for real-time RF signal classification on RISC-V, using compile-time tensor shapes and mdspan to enable compiler **inlining, loop unrolling, and vectorization** of matrix operations.
- Designed an expression system to **combine multiple tensor operations into a single computation**, reducing intermediate calculations and enabling SIMD vectorization.
- Implemented a custom memory allocator that uses a fixed memory buffer instead of repeated dynamic allocations, reducing runtime overhead on resource-constrained hardware.

FinSight AI | *Python, JavaScript, Next.js, Flask, Supabase*

Apr 2025 – Jun 2025

- Developed machine learning models for **financial forecasting, anomaly detection, and correlation testing**, integrating results into interactive dashboards for analyzing personal spending trends.
- Coordinated with the client to communicate project progress and translate feedback into development priorities.

Machine Learning Research Projects | *Python, PyTorch, Matplotlib, Hugging Face*

Mar 2025 – Jun 2025

- Object Detection with YOLOv1: Increased mAP score by 10%**, stabilized training, and reduced training time by modifying model architecture (bounding boxes, hyperparameters, dynamic learning rate).
- Movie Reviews Sentiment Analysis:** Fine-tuned DistilBERT using head-only training and LoRA to compare tradeoffs in training time, memory usage, and accuracy.
- DCGAN Optimization for Image Generation:** Improved DCGAN image quality and training stability through data augmentation, hyperparameter tuning, and architectural changes to prevent model collapse.

SOLPower | *Python, Flask, React, JavaScript, D3.js, XGBoost, Groq*

Feb 2025

- Led full-stack development for an **AI-powered solar energy site suitability web application** under tight deadlines.
- Built an interactive map of the United States with D3.js and integrated conversational AI through Groq for additional insights.

Certifications / Awards

Certifications: Amazon Web Services Certified Cloud Practitioner

UCSB Datathon 2024 Second Place: Red wine quality calculator using AI model to sample different ingredient combinations.

UCSB Datathon 2025 First Place: Interactive AI-powered solar infrastructure optimization tool with chatbot features.